

```

1  /* =====
2  GENERATE EMI SCHEDULE
3  ===== */
4  CREATE OR ALTER PROCEDURE Generate_EMI_Schedule
5  (
6      @LoanID INT
7  )
8  AS
9  BEGIN
10     SET NOCOUNT ON;
11
12     BEGIN TRY
13         BEGIN TRANSACTION;
14
15         -- 1. Validate loan
16         IF NOT EXISTS
17         (
18             SELECT 1
19             FROM Loans
20             WHERE LoanID = @LoanID
21         )
22             THROW 50401, 'Loan does not exist.', 1;
23         END;
24
25         -- 2. Get loan details
26         DECLARE
27             @LoanAmount DECIMAL(18,2),
28             @InterestRate DECIMAL(5,2),
29             @TenureMonths INT,
30             @EMIAmount DECIMAL(18,2),
31             @DisbursementDate DATE,
32             @LoanStatus VARCHAR(15);
33
34         SELECT
35             @LoanAmount = LoanAmount,
36             @InterestRate = InterestRate,
37             @TenureMonths = LoanTenureMonths,
38             @EMIAmount = EMIAmount,
39             @DisbursementDate = DisbursementDate,
40             @LoanStatus = LoanStatus
41         FROM Loans WITH (UPDLOCK, HOLDLOCK)
42         WHERE LoanID = @LoanID;
43
44         -- 3. Loan must be disbursed
45         IF @LoanStatus <> 'Disbursed'
46             BEGIN
47                 THROW 50402, 'EMI schedule can be generated only for a disbursed loan.', 1;
48             END;
49
50         -- 4. Loan must have a disbursement date
51         IF @DisbursementDate IS NULL
52             BEGIN
53                 THROW 50403, 'Loan has not been disbursed.', 1;
54             END;
55
56         -- 5. Prevent duplicate EMI schedule
57         IF EXISTS
58         (
59             SELECT 1
60             FROM EMICollections
61             WHERE LoanID = @LoanID
62         )
63             BEGIN
64                 THROW 50404, 'EMI schedule already exists for this loan.', 1;
65             END;
66     END TRY
67     BEGIN CATCH
68         IF @@TRANCOUNT > 0 ROLLBACK TRANSACTION;
69         THROW;
70     END CATCH;
71     GO

```

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-- 6. Calculate EMI
SET @EMIAmount =
    dbo.Calculate_EMI
    (
        @LoanAmount,
        @InterestRate,
        @TenureMonths
    );

-- 7. Store calculated EMI in Loans table
UPDATE Loans
SET
    EMIAmount = @EMIAmount
WHERE LoanID = @LoanID;

-- 8. Variables for amortization
DECLARE
    @InstallmentNumber INT = 1,
    @OpeningPrincipal DECIMAL(18,2) = @LoanAmount,
    @InterestAmount DECIMAL(18,2),
    @PrincipalAmount DECIMAL(18,2),
    @ClosingPrincipal DECIMAL(18,2),
    @DueDate DATE;

-- 9. Generate EMI schedule
WHILE @InstallmentNumber <= @TenureMonths
BEGIN
    -- Due date for this installment
    SET @DueDate = DATEADD(MONTH, @InstallmentNumber, @DisbursementDate);

    -- Calculate monthly interest
    SET @InterestAmount =
        ROUND(@OpeningPrincipal * (@InterestRate / 100.0) / 12, 2);

    -- Principal portion
    SET @PrincipalAmount = @EMIAmount - @InterestAmount;

    -- Adjust if principal exceeds remaining
    IF @PrincipalAmount > @OpeningPrincipal
        BEGIN
            SET @PrincipalAmount = @OpeningPrincipal;
            SET @EMIAmount = @PrincipalAmount + @InterestAmount;
        END;

    -- Remaining principal
    SET @ClosingPrincipal = @OpeningPrincipal - @PrincipalAmount;

    -- Handle rounding issues
    IF ABS(@ClosingPrincipal) < 0.01
        SET @ClosingPrincipal = 0.00;

    -- Insert EMI record
    INSERT INTO EMICollections
    (
        LoanID, InstallmentNumber, DueDate, PaidDate, AmountDue,
        AmountPaid, PenaltyAmount, PaymentStatus, TransactionID,
        PrincipalAmount, InterestAmount, OutstandingPrincipal
    )
    VALUES
    (
        @LoanID, @InstallmentNumber, @DueDate, NULL, @EMIAmount,
        0.00, 0.00, 'Pending', NULL,
        @PrincipalAmount, @InterestAmount, @ClosingPrincipal
    );

    -- Next month
    SET @OpeningPrincipal = @ClosingPrincipal;
    SET @InstallmentNumber = @InstallmentNumber + 1;

    -- 10. Synchronize outstanding amount
    UPDATE Loans
    SET OutstandingAmount = @LoanAmount
    WHERE LoanID = @LoanID;

    COMMIT TRANSACTION;
    SELECT 'EMI schedule generated successfully.' AS Message;
END TRY
BEGIN CATCH
    IF @@TRANCOUNT > 0 ROLLBACK TRANSACTION;
    THROW;
END CATCH;
GO

```

```

1  /* =====
2  COLLECT EMI PAYMENT
3  ===== */
4  CREATE OR ALTER PROCEDURE Collect_EMI_Payment
5  (
6      @LoanID INT,
7      @InstallmentNumber INT,
8      @PaymentAmount DECIMAL(18,2),
9      @PaymentMode VARCHAR(20) = 'Branch',
10     @TransactionID INT = NULL
11 )
12 AS
13 BEGIN
14     SET NOCOUNT ON;
15
16     BEGIN TRY
17         BEGIN TRANSACTION;
18
19         -- 1. Validate installment record
20         IF NOT EXISTS
21         (
22             SELECT 1
23             FROM EMICollections
24             WHERE LoanID = @LoanID
25                 AND InstallmentNumber = @InstallmentNumber
26                 AND PaymentStatus = 'Pending'
27         )
28             THROW 50501, 'Pending installment not found.', 1;
29         END;
30
31         -- 2. Declare variables
32         DECLARE
33             @AmountDue DECIMAL(18,2),
34             @PrincipalAmount DECIMAL(18,2),
35             @InterestAmount DECIMAL(18,2),
36             @PenaltyAmount DECIMAL(18,2),
37             @DueDate DATE,
38             @PaidDate DATE = CAST(SYSDATETIME() AS DATE);
39
40         -- 3. Get installment details
41         SELECT
42             @AmountDue = AmountDue,
43             @PrincipalAmount = PrincipalAmount,
44             @InterestAmount = InterestAmount,
45             @PenaltyAmount = PenaltyAmount,
46             @DueDate = DueDate
47         FROM EMICollections WITH (UPDLOCK, ROWLOCK)
48         WHERE LoanID = @LoanID
49             AND InstallmentNumber = @InstallmentNumber
50             AND PaymentStatus = 'Pending';
51
52         -- 4. Calculate penalty if overdue
53         IF @DueDate < @PaidDate
54             BEGIN
55                 DECLARE @DaysLate INT = DATEDIFF(DAY, @DueDate, @PaidDate);
56                 SET @PenaltyAmount = ROUND(@AmountDue * 0.02 * @DaysLate, 2);
57             END
58         ELSE
59             BEGIN
60                 SET @PenaltyAmount = 0.00;
61             END;
62     END TRY
63     BEGIN CATCH
64         SET @PenaltyAmount = 0.00;
65     END;
66     GO

```

```

-- 5. Validate payment amount
IF @PaymentAmount < (@AmountDue + @PenaltyAmount)
BEGIN
    THROW 50502, 'Payment amount is less than total due amount.', 1;
END;

-- 6. Update installment record
UPDATE EMICollections
SET
    PaidDate = @PaidDate,
    AmountPaid = @PaymentAmount,
    PenaltyAmount = @PenaltyAmount,
    PaymentStatus = 'Paid',
    TransactionID = @TransactionID
WHERE LoanID = @LoanID
    AND InstallmentNumber = @InstallmentNumber;

-- 7. Update loan outstanding
UPDATE Loans
SET
    OutstandingAmount = OutstandingAmount - @PrincipalAmount,
    ModifiedDate = SYSDATETIME()
WHERE LoanID = @LoanID;

-- 8. Record transaction
IF @TransactionID IS NULL
BEGIN
    INSERT INTO Transactions
    (
        AccountID,
        TransactionType,
        TransactionCategory,
        PaymentChannel,
        Amount,
        Charges,
        TaxAmount,
        TransactionStatus,
        Remarks,
        TransactionDate,
        CreatedDate
    )
    VALUES
    (
        (SELECT AccountID FROM Loans WHERE LoanID = @LoanID),
        'Debit',
        'EMI Payment',
        @PaymentMode,
        @PaymentAmount,
        0.00,
        0.00,
        'Success',
        'EMI payment collected for LoanID ' + CAST(@LoanID AS VARCHAR(20)),
        SYSDATETIME(),
        SYSDATETIME()
    );
END;

COMMIT TRANSACTION;
SELECT 'EMI payment collected successfully.' AS Message;
END TRY
BEGIN CATCH
    IF @@TRANCOUNT > 0 ROLLBACK TRANSACTION;
    THROW;
END CATCH;
GO

```

```
--Example : Generate EMI schedule
EXEC Generate_EMI_Schedule
    @LoanID = 21;
--Example : View generated EMI collections
select * from EMICollections
--Example : View loan details
select * from Loans
where LoanID = 21
--Example : View updated account balance
select * from Accounts
where AccountID=34
--Example : View EMI transaction
select * from Transactions
where AccountID=34
```

92 %

Results Messages

	EMIID	LoanID	InstallmentNumber	DueDate	PaidDate	AmountDue	AmountPaid	PenaltyAmount	PaymentStatus	TransactionID	PrincipalAmount	InterestAmount	OutstandingPrincipal
1	73	21	1	2024-08-27	NULL	47851.28	0.00	0.00	Pending	NULL	44404.08	3447.20	507148.59
2	74	21	2	2024-09-27	NULL	47851.28	0.00	0.00	Pending	NULL	44681.60	3169.68	462466.99
3	75	21	3	2024-10-27	NULL	47851.28	0.00	0.00	Pending	NULL	44960.86	2890.42	417506.13
4	76	21	4	2024-11-27	NULL	47851.28	0.00	0.00	Pending	NULL	45241.87	2609.41	372264.26
5	77	21	5	2024-12-27	NULL	47851.28	0.00	0.00	Pending	NULL	45524.63	2326.65	326739.63
6	78	21	6	2025-01-27	NULL	47851.28	0.00	0.00	Pending	NULL	45809.16	2042.12	280930.47
7	79	21	7	2025-02-27	NULL	47851.28	0.00	0.00	Pending	NULL	46095.46	1755.82	234835.01
8	80	21	8	2025-03-27	NULL	47851.28	0.00	0.00	Pending	NULL	46383.56	1467.72	188451.45
9	81	21	9	2025-04-27	NULL	47851.28	0.00	0.00	Pending	NULL	46673.46	1177.82	141777.99
10	82	21	10	2025-05-27	NULL	47851.28	0.00	0.00	Pending	NULL	46965.17	886.11	94812.82
11	83	21	11	2025-06-27	NULL	47851.28	0.00	0.00	Pending	NULL	47258.70	592.58	47554.12
12	84	21	12	2025-07-27	NULL	47851.28	0.00	0.00	Pending	NULL	47554.07	297.21	0.05

 **Generate EMI Schedule**

	LoanID	LoanAccountNumber	CustomerID	AccountID	LoanType	LoanAmount	InterestRate	LoanTenureMonths	EMIAmount	LoanStatus	ApplicationDate	ApprovalDate	DisbursementDate	ApprovedByEmployeeID	OutstandingAmount	Purpose
1	21	LN007000021	31	34	Gold	551552.67	7.50	12	47851.28	Disbursed	2024-08-01	2024-07-19	2024-07-27	5	551552.67	Working capital against gold jewellery

	AccountID	AccountNumber	CustomerID	BranchID	AccountTypeID	OpeningDate	CurrentBalance	AvailableBalance	LienAmount	OverdraftLimit	CurrencyCode	AccountStatus	FreezeReason	ClosedDate	ClosedByEmployeeID	LastTransactionDate	CreatedDate	ModifiedDate
1	34	BSGB007000034	31	7	2	2023-06-08	317404.72	317404.72	0.00	0.00	INR	Active	NULL	NULL	NULL	2026-07-06 00:00:00.0000000	2026-08-09 17:07:30.1766834	NULL

	TransactionID	TransactionReferenceNumber	TransferReferenceNumber	AccountID	TransactionType	TransactionCategory	PaymentChannel	Amount	Charges	TaxAmount	OpeningBalance	ClosingBalance	CounterpartyAccountNumber	CounterpartyName	CounterpartyBank	TransactionStatus	Remarks
--	---------------	----------------------------	-------------------------	-----------	-----------------	---------------------	----------------	--------	---------	-----------	----------------	----------------	---------------------------	------------------	------------------	-------------------	---------


```
-- Collect EMI installment 1
EXEC Collect_EMI
    @EMIID = 73
-- Collect EMI installment 2
EXEC Collect_EMI
    @EMIID = 74
-- Collect EMI installment 3
EXEC Collect_EMI
    @EMIID = 75
--Example : View generated EMI collections
select * from EMICollections
--Example : View loan details
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92 %

Results Messages

	EMIID	LoanID	InstallmentNumber	DueDate	PaidDate	AmountDue	AmountPaid	PenaltyAmount	PaymentStatus	TransactionID	PrincipalAmount	InterestAmount	OutstandingPrincipal
1	73	21	1	2024-08-27	2026-08-16	47851.28	47851.28	0.00	Paid	58	44404.08	3447.20	507148.59
2	74	21	2	2024-09-27	2026-08-16	47851.28	47851.28	0.00	Paid	59	44681.60	3169.68	462466.99
3	75	21	3	2024-10-27	2026-08-16	47851.28	47851.28	0.00	Paid	60	44960.86	2890.42	417506.13
4	76	21	4	2024-11-27	NULL	47851.28	0.00	0.00	Pending	NULL	45241.87	2609.41	372264.26
5	77	21	5	2024-12-27	NULL	47851.28	0.00	0.00	Pending	NULL	45524.63	2326.65	326739.63
6	78	21	6	2025-01-27	NULL	47851.28	0.00	0.00	Pending	NULL	45809.16	2042.12	280930.47
7	79	21	7	2025-02-27	NULL	47851.28	0.00	0.00	Pending	NULL	46095.46	1755.82	234835.01
8	80	21	8	2025-03-27	NULL	47851.28	0.00	0.00	Pending	NULL	46383.56	1467.72	188451.45
9	81	21	9	2025-04-27	NULL	47851.28	0.00	0.00	Pending	NULL	46673.46	1177.82	141777.99
10	82	21	10	2025-05-27	NULL	47851.28	0.00	0.00	Pending	NULL	46965.17	886.11	94812.82
11	83	21	11	2025-06-27	NULL	47851.28	0.00	0.00	Pending	NULL	47258.70	592.58	47554.12
12	84	21	12	2025-07-27	NULL	47851.28	0.00	0.00	Pending	NULL	47554.07	297.21	0.05



Collect
EMI Schedule

	LoanID	LoanAccountNumber	CustomerID	AccountID	LoanType	LoanAmount	InterestRate	LoanTenureMonths	EMIAmount	LoanStatus	ApplicationDate	ApprovalDate	DisbursementDate	ApprovedByEmployeeID	OutstandingAmount	Purpose
1	21	LN007000021	31	34	Gold	551552.67	7.50	12	47851.28	Disbursed	2024-08-01	2024-07-19	2024-07-27	5	417506.13	Working capital against gold jewellery

	AccountID	AccountNumber	CustomerID	BranchID	AccountTypeID	OpeningDate	CurrentBalance	AvailableBalance	LienAmount	OverdraftLimit	CurrencyCode	AccountStatus	FreezeReason	ClosedDate	ClosedByEmployeeID	LastTransactionDate	CreatedDate	ModifiedDate
1	34	BSGB007000034	31	7	2	2023-06-08	173850.88	173850.88	0.00	0.00	INR	Active	NULL	NULL	NULL	2026-08-16 18:54:48.4572174	2026-08-09 17:07:30.1766834	2026-08-16

	TransactionID	TransactionReferenceNumber	TransferReferenceNumber	AccountID	TransactionType	TransactionCategory	PaymentChannel	Amount	Charges	TaxAmount	OpeningBalance	ClosingBalance	CounterpartyAccountNumber	CounterpartyName	CounterpartyBank	TransactionStatus	Remark
1	58	EMI-20260816-000056	NULL	34	Debit	Loan EMI	Auto Debit	47851.28	0.00	0.00	317404.72	269553.44	NULL	NULL	NULL	Success	EMI pa
2	59	EMI-20260816-000057	NULL	34	Debit	Loan EMI	Auto Debit	47851.28	0.00	0.00	269553.44	221702.16	NULL	NULL	NULL	Success	EMI pa
3	60	EMI-20260816-000058	NULL	34	Debit	Loan EMI	Auto Debit	47851.28	0.00	0.00	221702.16	173850.88	NULL	NULL	NULL	Success	EMI pa

Query executed successfully.

SATYAM\SQLXPRESS (16.0 RTM) SATYAM\ABHI (67) BankingSystem 00:00:00 17 rows